

AEQUITAS NODE OPERATOR GUIDE

Version 1.0 · June 2026 · aequitas.digital

Complete step-by-step guide · No prior blockchain experience required · ~20-30 min

What is an Aequitas Node?

An Aequitas node runs in the cloud and participates in the network. It validates human registrations, produces blocks, and keeps the blockchain alive. Node operators earn a daily share of all protocol fees — automatically at **20:00 Berlin time (CEST/CET)**.

Before You Start — What You Need

1. **Aequitas account:** Register via the Android app. You need a wallet address to receive rewards.
2. **GitHub account (free):** github.com — needed to fork the Aequitas code.
3. **Railway account (free):** railway.app, sign in with GitHub. No server needed.
4. **Dedicated node wallet (MetaMask):** A separate wallet just for your node. Export its private key: MetaMask → Account Details → Show Private Key. Keep this strictly private.
5. **10-30 minutes** — Railway does most of the work automatically.

Environment Variables — Complete Reference

SECURITY: Your `RELAYER_PRIVATE_KEY` is like a master password. Never share it.

Variable	Required?	What to set
<code>DATABASE_URL</code>	YES	Auto-injected by Railway when PostgreSQL is in the same project.
<code>RELAYER_PRIVATE_KEY</code>	YES	Private key of your node wallet (0x..., 66 chars). MetaMask → Account Details → Show Private Key.
<code>RELAYER_ADDRESS</code>	Recommended	Public address matching <code>RELAYER_PRIVATE_KEY</code> (0x..., 42 chars).
<code>NODE_OPERATOR_WALLET</code>	For rewards	Your Aequitas HUMAN wallet. Receives daily validator rewards at 20:00 Berlin.
<code>PEER_SECRET</code>	Multi-node	Shared secret — ALL nodes must use the SAME value. Get from network operator.
<code>SELF_URL</code>	Multi-node	Your node public URL: <code>https://YOUR-NAME.up.railway.app</code>
<code>PRIMARY_NODE_URL</code>	Multi-node	Set to: <code>https://aequitas.digital</code>
<code>NODE_KEY</code>	Optional	Base64 libp2p key for stable peer ID. If unset: auto-generated, printed to stderr as "SAVE THIS AS NODE_KEY ENVIRONMENT VAR: ". Copy and set it.
<code>IS_PRIMARY_NODE</code>	NO	Leave false. Only the official primary node uses true.
<code>RESET_STATE</code>	NO	DANGEROUS: wipes database on restart. Development only.

Step-by-Step Deployment on Railway

Step 1 — Fork Repository

Open github.com/hanoi96international-gif/Aequitas → click **Fork** → **Create fork**.

Step 2 — Create PostgreSQL Database

railway.app → **New Project** → **+ New** → **Database** → **Add PostgreSQL**. Railway auto-injects `DATABASE_URL`.

Step 3 — Deploy Node

In same project: **+ New** → **GitHub Repo** → select your fork → **Deploy Now**.

Step 4 — Set Variables

Service → **Variables**. Set: RELAYER_PRIVATE_KEY, RELAYER_ADDRESS, NODE_OPERATOR_WALLET, PEER_SECRET, SELF_URL, PRIMARY_NODE_URL=https://aequitas.digital. Save → auto-redeploy.

Step 5 — Get Public URL

Settings → Networking → **Generate Domain**. Open https://YOUR-URL/api/status — height should climb.

Step 6 — Verify Logs

Look for: "API Server listening on port 8080", "[SYNC] Connected to peer", "[NODE] Registered node operator wallet".

Step 7 — Earn Rewards

Rewards distributed automatically every day at **20:00 Berlin time**. 40% of all protocol fees split among node operators. Just keep your node running.

Troubleshooting

Symptom	Likely Cause	Solution
Height stays at 0	PRIMARY_NODE_URL missing	Set PRIMARY_NODE_URL=https://aequitas.digital + SELF_URL. Redeploy.
DATABASE_URL error	Wrong connection string	Format: postgres://user:pass@host:5432/dbname
"no code at address"	V7 not deployed yet	Normal on first start — node auto-deploys. Wait a few seconds.
No rewards	NODE_OPERATOR_WALLET missing	Add NODE_OPERATOR_WALLET=0xYOUR_HUMAN_WALLET.
"Application error" (Railway)	Build/startup failure	Check Deploy Logs. Common: missing DATABASE_URL or wrong key format.

Questions: github.com/hanoi96international-gif/Aequitas · aequitas.digital · UBI: 20:00 Berlin time daily